

Asma Jahan, MBBS, MHI

Data Analyst

Charlotte, NC • drasmakhan727@gmail.com • (980)-327-2103

PROFESSIONAL SUMMARY

Clinical Data Analyst with an MBBS and a Master's in Health Informatics & Analytics, experienced in working with EHR data, public health datasets, population health indicators, and research analytics. Strong analytical skills in SQL, Python, ETL, and clinical workflow analysis with hands-on experience supporting quality improvement, clinical operations, and academic research. Demonstrated ability to partner with clinicians, researchers, and administrators to improve data quality and support evidence-based care within healthcare environments.

CORE SKILLS

Healthcare & EHR Data: Epic, REDCap, clinical workflows, HIS data, data quality audits

Clinical Coding & Documentation: ICD-10-CM/PCS codes, AHA coding clinic standards.

Interoperability Standards: HL7 v2/v3, FHIR, CDA, USCDI, message parsing, data exchange troubleshooting

Analytics: SQL, Python, Pandas, advanced Excel, SAS, quality metric reporting, KPIs

Data Engineering: ETL pipelines, SQL Server, PostgreSQL, AWS (Lambda, Glue, S3)

BI Reporting: Tableau, Power BI, dashboards, operational & research reporting

Healthcare Standards: HIPAA, IRB, clinical documentation, regulatory compliance, NCQA-HEDIS.

Domains: Population health analytics, quality improvement, research informatics

Soft Skills: Communication, clinician partnership, problem solving, stakeholder alignment

PROFESSIONAL EXPERIENCE

Developer (Contract) - NMD Employer- Charlotte, NC | 2024 – Present

- Developed scalable ETL pipelines and automated workflows using AWS Lambda, Glue, and Kafka—transferable to healthcare data integration systems.
- Led SQL Server to AWS PostgreSQL migration for analytics modernization.
- Conducted data validation, schema modeling, and quality checks ensuring high integrity of enterprise data.
- Collaborated with cross-functional teams to improve reporting, governance, and data accessibility.

Graduate Research & Teaching Assistant — UNC Charlotte.

Charlotte, NC | Jan 2023 – May 2024

- Conducted analytics for health informatics and public health research using large public health datasets.

- Cleaned, transformed, and analyzed structured healthcare data using Python, SQL, and visualization tools.
- Prepared statistical summaries, analytical reports, and presentations for faculty and research teams.
- Assisted in teaching courses on database systems, clinical analytics, and healthcare data management.

Trial Management Intern — Repare Therapeutics - Remote | May 2023 – Aug 2023

- Supported clinical trial operations including data review, protocol documentation, and query resolution.
- Ensured high-quality data capture for oncology research studies, strengthening research informatics foundation.

Clinical Data Analyst — Aswini Hospitals - India | Nov 2020 – Nov 2022

- Performed **clinical data analysis using complex SQL queries** to assess **data coverage and completeness** across multiple healthcare domains.
- Designed and executed **domain-level coverage metrics**, collaborating with clinical subject-matter experts to define **numerator and denominator logic**.
- Analyzed **medical terminologies** including **ICD, CPT, SNOMED CT, LOINC, RxNorm, and NDC** to identify content gaps and inconsistencies.
- Supported **terminology mapping and hierarchical analysis**, ensuring accurate representation of clinical concepts within relational data models.
- Worked with **EHR-derived datasets** to validate clinical documentation, coding accuracy, and terminology completeness.
- Conducted **trend analysis and reporting** to track improvements in terminology coverage over time.
- Evaluated **external clinical data sources** (e.g., clinical trial data, research datasets) to identify missing concepts, inclusion/exclusion criteria gaps, and opportunities for content expansion..
- Contributed to **quality and process improvement initiatives** by documenting workflows, assumptions, and analytical methodologies.

NOTABLE PROJECTS

Hypertension Diagnostic Tool (Python): Developed a rule-based program to identify hypertension and triggers for physician consultation.

CMS Medicare Provider Analytics (SAS): Performed demographic and utilization analysis using t-tests, linear and logistic regression.

HL7/FHIR Claims Denial Model: Designed a claims denial monitoring workflow using interoperability standards.

EDUCATION & CERTIFICATIONS

Master of Science in Health Informatics & Analytics — UNC Charlotte, NC

Bachelor of Medicine, Bachelor of Surgery (MBBS) — KBN Medical Sciences, India

Certifications: CITI IRB (Social & Behavioral, Data/Specimens), CITI GCP, Telehealth Certification (AHA), Licensed Medical Doctor (India)

RESEARCH EXPERIENCE

Cost-effectiveness of Nivolumab vs. Traditional Chemotherapy in advanced NSCLC.

Chronic Alcoholism as a Risk Factor in the development of Type 2 Diabetes Mellitus.
